Lucas Di Salvo

About Me

I'm a student and teaching assistant in the Computer Science department at the Faculty of Exact and Natural Sciences of the University of Buenos Aires (FCEN - UBA). I consider myself as a very proactive person that is interested in many topics in computer science, while also doing a deep dive on various concepts regarding programming languages. This has led me to pursue an academic career, and share this interest of mine through teaching, a job that I like very much.

Personal Links

LinkedIn | GitHub | ldisalvo@dc.uba.ar

Education

2017 - ongoing **MSc. in Computer science**
Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires

Thesis (In progress since January 2026): Extending a reversible operational semantics for imperative programming languages with concurrency
- With Hernán Melgratti as my supervisor.

Elective courses taken[1] (15/12 points, 12 needed and 3 extra points):

- Behavioral Types and Contracts (64 hs)
  Binary session types, first order contracts, high-order contracts, session types cotracts, multiparty session types, multiparty session types contracts and dependent session types.
- Functional Approach to EDSLs (15 hs) (ECI2024)
- The Lambda-Calculus:
  From Simple Types to Non-Idempotent Intersection Types (15 hs) (ECI2024)
- Rewriting, Lambda-Calculus and Types (64 hs)
  Abstract rewritting systems, term rewritting systems, untyped lambda-calculus, normalization strategies, simply-typed lambda-calculus, explicit substitutions.
- Concurrent Programming (64 hs)
  Concurrency basics, synchronization mechanisms, shared memory models and properties, concurrent data structures, Software Transactional Memory.
- Linear Logic: Syntax, Models and Extensions (15 hs) (ECI2023)
- Heterogeneous programming models:
  Portability, Performance and Programmability (15 hs) (ECI2022)
- Introduction to APIs (24 hs)

Mandatory courses taken[1] (20/23):

- Software Engineering II (160 hs)
  Dynamic analysis of programs (genetic and evolutionary algorithms, fuzzing, etc.), static analysis of programs (dataflow, interprocedural, intraprocedural, etc.), software verification.
- Numerical Methods (240 hs)
- Operating Systems (160 hs)
  Unix and Bash, Processes and SO API, Scheduling, synchronization and concurrency mechanisms, memory management, drivers and I/O, file systems, security and virtualization.
- Software Engineering I (160 hs)
  OOP in SmallTalk, TDD and design principles.
- Language Theory (120 hs)
  Automata theory, regular expressions, grammars, parsers.
- Computer Organization II (160 hs)

[1] The hour count only accounts for the in-class time.

Intel architecture, ASM introduction, ASM and C ABI, SIMD, microarchitecture, cache, paging and segmentation, memory optimization techiniques.

- Programming Language Paradigms (120 hs)
  Functional programming in Haskell, Lambda-calculus, type inference, subtyping, resolution, logical programming in Prolog, classification and prototyping in Javascript.
- Algorithms and Data Structures III (240 hs)
  Graph theory, backtracking, dynamic programming, greedy algorithms, heuristics and metaheuristics, shortest pat, MST, max-flow, introduction to complexity theory, SAT. Done in C/C++ for algorithm design, and Python for dataset preparation, experimenting and analysis.
- Logic and Computability (112 hs)
  Turing machines, Halting problem, propositional logic, first order logic.
- Algorithms and Data Structures II (240 hs)
  Formal specification of abstract data types, trees and complex data structures, search and sorting algorithms, asymptotic analysis of algorithmis. Done in C/C++.
- Computer Organization I (112 hs)
  Von Neumann architecture, data representation, logic circuit design, CPU architecture, introduction to microarchitecture, memory, I/O, interruptions and buses.
- Algorithms and Data Structures I (240 hs)
  Formal specification of programs, weakest precondition, Hoare logic, general programming concepts, testing. Done in C/C++.
- Probability and Statistics (160 hs)
- Calculus II (160 hs)
- Algebra I (208 hs)
- CBC (*Common Basic Cycle*, a one year entrance program to Buenos Aires University)
  - Calculus I, Algebra, Introduction to Scientific Thinking, Introduction to Knowledge of Society and State, Chemistry, Physics.

2011 - 2016 **Técnico en Computación** (Technical high-school in programming & IT)
E.T. N°3 “María Sanchez de Thompson”

Academic Experience

All positions were obtained by a public selection and ranking process.

2024 - ongoing
Teaching Assistant of “Programming Language Paradigms” at FCEN - UBA
Preparation and lecturing of practical classes for hundreds of students. Exams and projects writting, assessment and evaluation, among others. Main topics covered: Functional programming in Haskell, equational reasoning, natural deduction (propositional logic and first order logic), Lambda-calculus, type inference, resolution, logical programming in Prolog, OOP in SmallTalk.
- 2024 results 2-year position.
- 2023 results Annual position.

2025 - 2026 (1 year)
Teacher of LaTeXworkshop at FCEN - UBA
Creation and lecturing of a monthly 3-hour LaTeXworkshop.
The aim of the workshop is to give tools to new students to tackle written assignments, reports and such, as well as being able to take stylized and detailed notes.
- This position was an experimental project. After its success and usefulness, it is now a regular position with public selection.

August 2023 - February 2024
Teaching Assistant of “Logic and Computability” at FCEN - UBA
Preparation and lecturing of practical classes. Exams assessment and evaluation, among others. Main topics covered: Turing machines, Halting problem, propositional logic, first order logic.
- The results platform was not in use at that time.

2020
Computer science popularizer at FCEN - UBA
Preparation of courses for high school students about different Computer Science topics, stands in science fairs and vocational orientation talks.
- The results platform was not in use at that time.

2019 - 2021
Programming lessons (done independently)

Organization and teaching to high school students. The topics include but are not limited to: Python programming, OOP, structured programming, C#, Java, data structures and an introduction to algorithms.

Industry Experience

2021 - 2022 (7 months)	**Game Developer at "Globant"** Making of custom UIs and interactions for different game systems, such as a BattlePass, PopUps, Item selectors and more.
2019 (6 months)	**Software Developer at "Seincomp Informática"** Analysis, development and deployment of .NET (VB, C#, Microsoft SQL Server) customized applications, with its maintenance and support. Web applications development for internal usage, with JavaScript (Boostrap), HTML, CSS.

Science Popularization (Volunteering)

September 2024 September 2019	**Science Fair - Booth Speaker** Hosted a booth about algorithmic concepts like binary search, at the annual Computer science Week in my university. Made interesting through a pick and guess game, with high school students being the target audience.
November 2020	**Science Fair - Workshop Instructor** Hosted a workshop on Arduino and robotics for high school students, at the annual Computer Science Week in my university. This was made during Covid, so it was offered through Zoom and Tinkercad.
June 2018	**Science Fair - Guide** Guided high school classes through their schedule (registration, talks and venues) at the annual Computer Science Week in my university.

Research Activities

August 2022 - April 2025	Assisted LoReL (*Logic and Rewriting for Programming Languages*) seminar as a student. ◦ Topics covered vary, but mainly lambda-calculi, cathegory theory, linear logic and quantum computation.

Languages

Spanish	Native
English	Advanced (several years of formal education, used both for education and professionally)
Japanese	Basic comprehension (self taught)

Projects

- A project series for *Algorithms and Data Structures 2* (mandatory university course), designed to develop implementations for some of the most common data structures and its functionalities (programmed in C++).
  - A Doubly Linked List, a Binary Search Tree implemented on a *set*, a Map implemented on a *trie*, and a Priority Queue implemented on a *heap*.
- A small tool to fetch system information written in Haskell using System.IO to learn about monads (both in do-notation and *vanilla* monadic notation).

- A project series for *Algorithms and Data Structures 3* (mandatory university course), designed to research, analyze and develop algorithms using different programming techniques to address complex problems (programmed the core algorithms in C++, and Python3 using Jupyter notebook for the data preparation and analysis).
  - An analysis on the Subset Sum problem. The aim of this project was to ensure the correct procedure to develop solutions by using brute force, backtracking and dynamic programming for the problem at hand, and analyze the effectiveness and efficacy of said techniques, in great detail.
  - A project about developing heuristics over the Maximum Impact Coloring Polytope problem.
- A small EDSL for propositional logic with arithmetic comparison. Done as final project for "Functional approach to EDSLs" optional course.
- My resume (this document), made in LaTeX.